

Research Report

Research Query: synthetic data

Summary

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Sources and Content

Title: Byzantine-Resilient SGD in High Dimensions on Heterogeneous Data

URL: <http://arxiv.org/abs/2005.07866v1>

Abstract: We study distributed stochastic gradient descent (SGD) in the master-worker architecture under Byzantine attacks. We consider the heterogeneous data model, where different workers may have different local datasets, and we do not make any probabilistic assumptions on data generation. At the core of our algorithm, we use the polynomial-time outlier-filtering procedure for robust mean estimation proposed by Steinhardt et al. (ITCS 2018) to filter-out corrupt gradients. In order to be able to apply their filtering procedure in our heterogeneous data setting where workers compute stochastic gradients, we derive a new matrix concentration result, which may be of independent interest. We provide convergence analyses for smooth strongly-convex and non-convex objectives. We der

Title: Privacy Measurement in Tabular Synthetic Data: State of the Art and Future Research Directions

URL: <http://arxiv.org/abs/2311.17453v1>

Abstract: Synthetic data (SD) have garnered attention as a privacy enhancing technology. Unfortunately, there is no standard for quantifying their degree of privacy protection. In this paper, we discuss proposed quantification approaches. This contributes to the development of SD privacy standards; stimulates multi-disciplinary discussion; and helps SD researchers make informed modeling and evaluation decisions.

Title: Data Encoding for Byzantine-Resilient Distributed Optimization

URL: <http://arxiv.org/abs/1907.02664v2>

Abstract: We study distributed optimization in the presence of Byzantine adversaries, where both data and computation are distributed among m worker machines, t of which may be corrupt. The compromised nodes may collaboratively and arbitrarily deviate from their pre-specified programs, and a designated (master) node iteratively computes the model/parameter vector for generalized linear models. In this work, we primarily focus on two iterative algorithms: Proximal Gradient Descent (PGD) and Coordinate Descent (CD). Gradient descent (GD) is a special case of these algorithms. PGD is typically used in the data-parallel setting, where data is partitioned across different samples, whereas, CD is used in the model-parallelism setting, where data is partitioned across the parameter space.

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